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Management apparatus, management system, and management method

Abstract

A management apparatus specifies a first communication path requirement template corresponding to a first communication path requirement of a first application module and a second communication path requirement template corresponding to a second communication path requirement of a second application module, determines the available communication module included in the first communication path requirement template to be a first communication module and determines the available communication module included in the second communication path requirement template to be a second communication module, determines an arrangement destination of the first communication module to be a first arrangement destination, and determines an arrangement destination of the second communication module to be a second arrangement destination, and sets a communication path by arranging the first application module and the first communication module in the first arrangement destination and arranging the second application module and the second communication module in the second arrangement destination.

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Background/Summary

CLAIM OF PRIORITY

(1) The present application claims priority from Japanese patent application JP 2022-012703 filed on Jan. 31, 2022, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a management apparatus, a management system, and a management method for managing a module.

2. Description of the Related Art

(3) The fifth-generation mobile communication (5G) enables low-delay, broadband, and highly-reliable wireless communication. Under such circumstances, various existing networks among private networks can be integrated by introducing a low-delay, broadband, and highly-reliable

wireless network characterized by 5G, and it can be expected that costs of network construction and operation can be reduced. In particular, in a system in which a plurality of networks are mixed, such as a system including an operation technology (OT), an effect thereof is remarkable. Application of 5G to various fields has been studied. One example of an application destination thereof is a site network in manufacturing and distribution sites.

(4) In the manufacturing and distribution sites, there is a demand for countermeasures against labor shortage, productivity improvement, and quality improvement by introducing new digital transformation (DX) solutions such as real-time work instructions utilizing high-definition video analysis, device control that enables people and robots to perform cooperative works, and remote control with realistic feeling.

(5) By contrast, it has been studied to realize highly-reliable and flexible service provision including feedback control in a cyber-physical system by introducing a local 5G or a public 5G and a general-purpose edge processing server (also referred to as multi-access edge computing (MEC)) to integrate and aggregate a site network into a wireless network.

(6) As a result, it can be expected that an application related to business can be flexibly and quickly changed due to factors such as a supply and demand fluctuation in a supply chain, a process change by personnel adjustment in a company, and fine work adjustment caused by a change in a site environment. The application mentioned herein is a program for achieving a certain purpose by combining a plurality of modules that provide some data input and output services. A change in a configuration of the application is to arbitrarily change a combination and a deployment position of the plurality of modules. The change in the configuration of the application is implemented by an instruction from a cloud by using the above-described factors as triggers. A network connecting these modules is required to provide highly-reliable connectivity following a configuration change.

(7) JP 2020-140276 A below discloses a network requirement generation system that generates a network requirement applied to a service. The network requirement generation system includes a service requirement acquisition unit that includes a computer including an arithmetic device that executes predetermined arithmetic processing to realize each of the following functional units and a storage device accessible by the arithmetic device, and acquires a service requirement input by a user, a service requirement analysis unit that analyzes the input service requirement to generate a network requirement, and a network requirement control unit that creates a network setting content for a control device that constructs a network from the generated network requirement.

(8) JP 2020-502948 A below discloses a network gateway for routing a data flow through a plurality of network connections. The network gateway includes a plurality of network interfaces including a first network interface for transmitting data through a plurality of network connections, and at least one processor. The processor is configured to transmit a sequential burst of packets through the first network interface, generates a bandwidth of the first network interface based on a timestamp and a size of the packet recorded when the packet in the sequential burst of packets is received at a receiving node, and route a data flow of the sequential packets through the plurality of network connections based on the generated bandwidth of the first network interface.

SUMMARY OF THE INVENTION

(9) Problems in wireless networking of the site network corresponding to a strict application requirement required by the application in the OT will be described by taking a factory network as an example. In a current factory network, a plurality of networks such as a control network, an information network, and an office automation (OA) network are mixed according to a requirement of business. There are various requirements such as a communication delay and a communication utilization rate for the control device for each application.

(10) The system is flexible by the wireless networking of the factory network, but the dependence on a system engineer increases. When the system engineer manually sets or constructs a system or a network to satisfy these requirements, the system engineer cannot follow a fluctuation in supply and demand, a change in site environment, or the like, and does not scale as business.

(11) The network requirement generation system disclosed in JP 2020-140276 A does not disclose provision of highly-reliable communication connectivity required by the application by merely generating a setting of a network device. The network gateway disclosed in JP 2020-502948 A does not disclose a dynamic and flexible introduction method into the system.
(12) An object of the present invention is to realize a setting of a communication path suitable for a request.
(13) A management apparatus according to an aspect of the invention disclosed in the present application is a management apparatus that is able to communicate with a computer group. The apparatus includes a processor that executes a program, and a storage device that stores the program, The storage device stores an application module group, a communication module group, a communication path requirement that is a condition required for a communication path between an application module in the application module group and another application module as a communication partner of the application module, a communication path requirement target that designates the other application module, and a communication path requirement template group that includes a communication path requirement definition that defines a type and a condition of communication in the communication path requirement, an available communication module that defines a communication module available to the communication path requirement definition, and an available setting that defines a setting of communication available to the communication path requirement definition. The processor executes specification processing of specifying a first communication path requirement template corresponding to a first communication path requirement of a first application module from the communication path requirement template group, and specifying a second communication path requirement template corresponding to a second communication path requirement of a second application module designated as a communication partner of the first application module by the communication path requirement target from the communication path requirement template group, communication module determination processing of determining the available communication module included in the first communication path requirement template to be a first communication module available by the first application module, and determining the available communication module included in the second communication path requirement template to be a second communication module available by the second application module, communication module arrangement destination determination processing of determining an arrangement destination of the first communication module to be a first arrangement destination of the first application module in the computer group, and determining an arrangement destination of the second communication module to be a second arrangement destination of the second application module in the computer group, arrangement processing of arranging the first application module and the first communication module in the first arrangement destination and arranging the second application module and the second communication module in the second arrangement destination, and communication path setting processing of setting a communication path that connects the first application module and the second application module to be able to communicate via the first communication module and the second communication module.
(14) According to the representative embodiment of the present invention, it is possible to realize the setting of the communication path suitable for the request. Other objects, configurations, and effects will be made apparent in the following descriptions of the embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram illustrating a system configuration example of an edge operation management system;

(2) FIG. **2** is a block diagram illustrating a hardware configuration example of a computer;
(3) FIG. **3** is a block diagram illustrating a functional configuration example of an orchestration server;
(4) FIG. **4** is an explanatory diagram illustrating an example of an application;
(5) FIG. **5** is an explanatory diagram illustrating an example of a resource management table;
(6) FIG. **6** is an explanatory diagram illustrating an example of an application requirement table;
(7) FIG. **7** is an explanatory diagram illustrating an example of an application management table;
(8) FIG. **8** is an explanatory diagram illustrating an example of a communication module repository;
(9) FIG. **9** is an explanatory diagram illustrating an example of a communication path requirement template table;
(10) FIG. **10** is a block diagram illustrating a functional configuration example of a processing server and an IoT gateway;
(11) FIG. **11**A is an explanatory diagram illustrating an example of a communication path management table retained by an edge processing server in a factory LAN;
(12) FIG. **11**B is an explanatory diagram illustrating an example of a communication path management table retained by an edge processing server in a local 5G network;
(13) FIG. **11**C is an explanatory diagram illustrating an example of a communication path management table retained by the IoT gateway;
(14) FIG. **11**D is an explanatory diagram illustrating an example of a communication path management table retained by a cloud processing server;
(15) FIG. **12** is a sequence diagram illustrating an example of a deployment operation of a module;
(16) FIG. **13** is a block diagram illustrating a configuration example of a logical system after deployment of the module;
(17) FIG. **14** is an explanatory diagram illustrating a display screen example of the orchestration server;
(18) FIG. **15** is a sequence diagram illustrating an example of an undeployment operation of the module;
(19) FIG. **16** is a sequence diagram illustrating an example of a re-deployment operation of the module;
(20) FIG. **17**A is an explanatory diagram illustrating an example of a communication path management table after re-deployment retained by the edge processing server in a public 5G network;
(21) FIG. **17**B is an explanatory diagram illustrating an example of a communication path management table after re-deployment retained by the IoT gateway;
(22) FIG. **18** is a block diagram illustrating a configuration example of a logical system after re-deployment of the module; and
(23) FIG. **19** is an explanatory diagram illustrating an example of a display screen of the orchestration server after re-deployment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(24) Edge Operation Management System

(25) FIG. **1** is a block diagram illustrating a system configuration example of an edge operation management system. An edge operation management system **1** includes a cloud **101** and a factory system **5**. The cloud **101** and the factory system **5** are connected to be able to communicate via a public network **100**. The cloud **101** and the factory system **5** are connected to a public 5G network **40** to be able to communicate via a public network **100**.

(26) The public 5G network **40** includes a base station **41**, an edge processing server **24**E**2**, and a mobile network management server **44**. The base station **41**, the edge processing server **24**E**2**, and the mobile network management server **44** are connected to be able to communicate via a mobile core network **43**. The mobile network management server **44** is connected to the public network

100 to be able to communicate. The mobile network management server **44** manages communication between the public 5G network **40**, and an IoT gateway **13**, a local 5G network **20**, a factory local area network (LAN) **32**, and the cloud **101** outside the public 5G network **40**. The base station **41** can communicate with the mobile core network **43** via a mobile backhaul (MBH) **42**.

(27) The cloud **101** includes an orchestration server **102** and a cloud processing server **24**C**1**. The orchestration server **102** is a management apparatus that controls the cloud processing server **24**C**1**, the edge processing server **24**E**2**, an edge processing server **24**E**3**, and an edge processing server **24**E**4** to integrally set a virtualization environment and automate an operation of the virtualization environment.

(28) The factory system **5** includes the edge processing server **24**E**3**, the factory LAN **32**, an existing factory network **34**, a factory LAN management server **35**, the local 5G network **20**, a wireless LAN base station **31**, and a terminal **15**. The edge processing server **24**E**3**, a firewall **33**, the existing factory network **34**, the factory LAN management server **35**, the local 5G network **20**, and the wireless LAN base station **31** are connected to be able to communicate via the factory LAN **32**. The factory LAN management server **35** manages communication between the factory LAN **32**, and the IoT gateway **13**, the local 5G network **20**, the cloud **101**, and the public 5G network **40** outside the factory LAN **32**.

(29) The local 5G network **20** includes the edge processing server **24**E**4**, a mobile core apparatus **23**, the base station **21**, and an MBH **22**. The edge processing server **24**E**4**, the mobile core apparatus **23**, and the base station **21** are connected to be able to communicate via the MBH **22**. The mobile core apparatus **23** is a generic name of an apparatus that accommodates long term evolution (LTE) or 5G mobile communication, and is also referred to as an evolved packet core (EPC) or a 5G core (5GC). The mobile core apparatus **23** manages communication between the local 5G network **20**, and the IoT gateway **13**, the factory LAN **32**, the cloud **101**, and the public 5G network **40** outside the local 5G network **20**.

(30) The terminal **15** is a computer including a camera **10**, a control device **11**, a bus **12**, and the IoT gateway **13**. The camera **10**, the control device **11**, and the IoT gateway **13** are connected to be able to communicate via a bus **12**. The terminal **15** is connected to the base station **21**, the wireless LAN base station **31**, and the base station **41** to be able to communicate via the IoT gateway **13**.

(31) The camera **10** images a work machine such as a worker or a conveyor in the factory system **5**. The control device **11** is connected to a work machine to be controlled, and gives control data such as a rotation speed of a roller to the work machine to control an operation of the work machine.

(32) One terminal **15** is provided in the factory system **5**. In a case where a plurality of manufacturing lines are present in the factory system **5**, the terminal **15** is provided for each manufacturing line.

(33) In a case where the edge processing server **24**E**2**, the edge processing server **24**E**3**, and the edge processing server **24**E**4** are not distinguished, these servers are simply referred to as an edge processing server **24**E. In a case where the cloud processing server **24**C**1**, the edge processing server **24**E**2**, the edge processing server **24**E**3**, and the edge processing server **24**E**4** are not distinguished, these servers are simply referred to as a processing server **24**. The processing server **24** is a computer.

Hardware Configuration Example of Computer (Orchestration Server **102**, Processing Server **24**, IoT Gateway **13**)

(34) FIG. **2** is a block diagram illustrating a hardware configuration example of a computer. A computer **200** includes a processor **201**, a storage device **202**, an input device **203**, an output device **204**, and a communication interface (communication IF) **205**. The processor **201**, the storage device **202**, the input device **203**, the output device **204**, and the communication IF **205** are connected by a bus **206**. The processor **201** controls the computer **200**. The storage device **202** is a work area of the processor **201**. The storage device **202** is a non-transitory or transitory recording

medium that stores various programs and data. Examples of the storage device **202** include a read only memory (ROM), a random access memory (RAM), a hard disk drive (HDD), and a flash memory. The input device **203** inputs data. Examples of the input device **203** include a keyboard, a mouse, a touch panel, a numeric keypad, a scanner, a microphone, and a sensor. The output device **204** outputs data. Examples of the output device **204** include a display, a printer, and a speaker. The communication IF **205** is connected to a network to transmit and receive data. Hereinafter, embodiments according to the present invention will be described.

Functional Configuration Example of Orchestration Server **102**

(35) FIG. **3** is a block diagram illustrating a functional configuration example of the orchestration server **102**. The orchestration server **102** includes a metric information collection unit **300**, an application requirement acquisition unit **301**, a module arrangement determination unit **302**, a communication path requirement interpretation unit **303**, a module deployment control unit **304**, an E2E network quality control unit **305**, a resource management table **500**, an application requirement table **600**, an application management table **700**, a communication module repository **800**, and a communication path requirement template table **900**.

(36) Specifically, the metric information collection unit **300**, the application requirement acquisition unit **301**, the module arrangement determination unit **302**, the communication path requirement interpretation unit **303**, the module deployment control unit **304**, and the E2E network quality control unit **305** are realized by causing the processor **201** to execute a program stored in the storage device **202** illustrated in FIG. **2**, for example.

(37) The metric information collection unit **300**, the application requirement acquisition unit **301**, the module arrangement determination unit **302**, the communication path requirement interpretation unit **303**, the module deployment control unit **304**, and the E2E network quality control unit **305** are stored in the storage device **202**.

(38) The metric information collection unit **300** collects metric information from the processing server **24** and the IoT gateway **13**. The metric information is information on a distance of a path (channel) to a communication partner, and is specifically, for example, a delay ([ms]), a jitter ([ms]), a bandwidth ([Mbps]), and a packet error rate (PER). The metric information collection unit **300** may collect and update the metric information at regular time intervals.

(39) The application requirement acquisition unit **301** receives an input of an application requirement by receiving an input from an external computer or by an operation of an operator. The application requirement is a condition required for an application. The application is a software module group including an application module and a communication module. The application is retained in the cloud **101**. In a case where the application module and the communication module are not distinguished from each other, these modules are simply referred to as a module.

(40) The application module is a software module that processes data, and includes, for example, a learning module that executes machine learning, a data collection module that collects data, a video analysis module that analyzes video, a camera control module that controls the camera, and a control device control module that controls the control device.

(41) The communication module is a module that transmits and receives data, and includes, for example, a broadband communication module that performs packet duplication transfer in a plurality of paths, a highly-reliable communication module that performs aggregation of a plurality of paths, a jitter reduction communication module that absorbs a packet jitter, and a WAF communication module that functions as a web application firewall (WAF).

(42) The application requirement is defined for each module, and includes a condition of which modules are arranged where, what communication path requirement, and with whom. The communication path requirement is a condition required for the communication path, and includes a communication type such as control communication or media communication and a constraint condition thereof. The communication path is a communication channel between the module and a communication partner thereof.

(43) The module arrange determination unit **302** determines an arrangement destination of the module. The arrangement destination includes, for example, ANY (anywhere), a closed area, and a field area. The closed area is in a device in which the module is arranged, and the field area indicates that the device in which the module is arranged is able to communicate with a plurality of networks.
(44) The communication path requirement interpretation unit **303** interprets the communication path requirement. Specifically, for example, the communication path requirement interpretation unit **303** specifies a communication path requirement template corresponding to the communication path requirement of the arrangement destination of the module from the communication path requirement template table **900**, and determines the communication module corresponding to the specified communication path requirement template.
(45) The module deployment control unit **304** performs control such that the module is arranged at the arrangement destination of the module determined by the module arrangement determination unit **302**. Specifically, for example, the module deployment control unit **304** transmits the module to the arrangement destination of the module determined by the module arrangement determination unit **302**.
(46) The E2E network quality control unit **305** controls END-TO-END, that is, the quality of the network present between the module arranged at the arrangement destination and the communication partner thereof. Specifically, for example, the E2E network quality control unit **305** executes quality of a service (QOS) setting and a communication path setting. Qos is an existing technology that adjusts the order and amount of data to enable stable use of services on a network.
Application
(47) FIG. **4** is an explanatory diagram illustrating an example of an application. An application **400** includes a camera control module **401**, a control device control module **402**, a data collection module **403**, a video analysis module **404**, a primary learning module **405**, and a secondary learning module **406**.
(48) The camera control module **401** instructs the camera **10** to capture an image and receives video data captured by the camera **10**. The camera control module **401** transmits the video data to the data collection module **403** when the primary learning model is trained, and transmits the video data to the video analysis module when prediction using the primary learning model is performed.
(49) The control device control module **402** acquires the control data set as the control target from the control device. The control device control module **402** transmits the control data to the data collection module **403** when the primary learning model is trained, and transmits the control data to the video analysis module **404** when prediction using the primary learning model is performed.
(50) The control device control module **402** transfers real-time feedback control data from the video analysis module **404** to the control device. The control device **11** controls the control target based on the real-time feedback control data. Specifically, for example, in a case where the real-time feedback control data is data indicating that the control data and the video data on which the video analysis is performed are normal works, the control device **11** continues the control of the control target as it is. On the other hand, in a case where the real-time feedback control data is data indicating that the control data and the video data on which the video analysis is performed are abnormal works deviated from each other, the control data is corrected. For example, in a case where a conveyance speed of the conveyor exceeds an upper limit speed, the control device **11** executes control to lower the number of rotations of the roller.
(51) The data collection module **403** associates the video data with the control data by time, and transfers, as learning data, the data to the primary learning module.
(52) The primary learning module **405** executes machine learning by using the learning data from the data collection module and correct answer data for each preset time, and generates a learning model. The correct answer data is, for example, an identification label for identifying whether the work at this time is a normal work or a deviated work. The primary learning module **405** transmits

the generated primary learning model to the secondary learning module **406** and the video analysis module **404**.
(53) The video analysis module **404** executes video analysis by using the primary learning model. The video analysis module **404** inputs the video data from the camera control module **401** and the control data from the control device control module **402** to the primary learning model, calculates a prediction result indicating whether the work at this time is a normal work or a deviated work, and transmits, as the real-time feedback control data, the prediction result to the control device control module **402**.
(54) The secondary learning module **406** acquires the primary learning model from the primary learning module **405** and generates a secondary learning module. Specifically, for example, the secondary learning module **406** extracts only a feature common to a factory system group including the factory system **5** and another factory system, re-constructs a primary learning model, and generates the primary learning model as a secondary learning model as know-how.
(55) Table Group in Orchestration Server **102**
(56) Next, a table group in the orchestration server **102** will be specifically described.
(57) FIG. **5** is an explanatory diagram illustrating an example of the resource management table **500**. The resource management table **500** is a table for managing computer resources available by the terminal **15**. The resource management table **500** includes, as fields, a management identifier **501**, an IoT device **502**, a retainment interface **503**, an available computer resource **504**, a position **505**, and metric information **506**.
(58) The management identifier **501** is an identifier for managing the computer resources available by the IoT device **502**, and corresponds to the IoT device **502** on a one-to-one basis. The IoT device **502** defines an IoT device in the terminal **15**. In the example of the terminal **15**, the IoT device **502** is the IoT gateway **13**.
(59) The retainment interface **503** is a communication interface retained by the IoT device **502**. In the example of the terminal **15**, the IoT gateway **13** has an interface connectable to the base station **21** of the local 5G network **20**, the wireless LAN base station **31**, and the public 5G network **40**.
(60) The available computer resource **504** is a computer resource that is able to communicate with the IoT device **502** and is available by the application, and is, for example, a cloud processing server **24**C**1**, an edge processing server **24**E**2**, an edge processing server **24**E**3**, and an edge processing server **24**E**4**.
(61) The position **505** indicates a location of the computer resource where the available computer resource **504** is present. Since the resource management table **500** defines an entry in units of the management identifier **501** indicating the IoT gateway, the position **505** of the IoT device **502** is also retained although not illustrated.
(62) The metric information **506** is information on communication between the IoT device **502** and the available computer resource **504**, and includes, as subfields, via local 5G **507**, via wireless LAN **508**, and via public 5G **509**. The metric information **506** is recorded for each of the via local 5G **507**, the via wireless LAN **508**, and the via public 5G **509**. The metric information **506** is collected by the metric information collection unit **300** as described above.
(63) FIG. **6** is an explanatory diagram illustrating an example of the application requirement table **600**. The application requirement table **600** is a table that defines the application requirement. The application requirement table **600** includes, as fields, an application identifier **601**, an application module **602**, an arrangement position **603**, a communication path requirement target **604**, a communication path requirement **605**, and a communication path direction **606**.
(64) The application identifier **601** is identification information that uniquely specifies the application **400**. In this example, “App-A” is the application identifier **601** of the application **400**.
(65) The application module **602** is a software module that constructs the application specified by the application identifier **601**. In the example of the application **400**, the application module **602** is the camera control module **401**, the control device control module **402**, the data collection module

403, the video analysis module **404**, the primary learning module **405**, and the secondary learning module **406**.

(66) The arrangement position **603** defines a position where the application module **602** can be arranged. The communication path requirement target **604** is a target that communicates with the application module **602** according to the communication path requirement **605**, that is, a destination module in a case where the application module **602** is used as a transmission source. The communication path requirement **605** is a condition required for a communication path between the application modules **602**, and includes a communication type such as control communication or media communication and a constraint condition thereof.

(67) The communication path direction **606** defines a forward direction or a reverse direction as a direction of a communication path for an entry in which the communication path requirement **605** is present. The forward direction indicates a transmission direction from the application module **602** to the communication path requirement target **604**, and the reverse direction indicates a transmission direction of data from the communication path requirement target **604** to the application module **602**.

(68) FIG. **7** is an explanatory diagram illustrating an example of the application management table **700**. The application management table **700** is a table for managing a state after the application is arranged. The application management table **700** includes, as fields, a management identifier **501**, an application identifier **601**, a classification **701**, a module **702**, a module address **703**, and an arrangement position address **704**.

(69) The classification **701** is a classification obtained by dividing the application specified by the application identifier **601** by type. In the example of the application **400**, an application module and a communication module are included as the classification **701**. The camera control module **401**, the control device control module **402**, the data collection module **403**, the video analysis module **404**, the primary learning module **405**, and the secondary learning module **406** are classified into application modules. A broadband communication module and a highly-reliable communication module are classified into communication modules.

(70) The module **702** defines in detail each of the application modules **602** and the communication modules classified by the classification **701**.

(71) The module address **703** is an IP address at the arrangement destination of the module **702**. The module address **703** is assigned at the arrangement destination of the module **702**. The arrangement position address **704** is an address of the arrangement destination of the module **702**. In the example of FIG. **7**, "C1.cloud.jp" is an address of the cloud processing server **24**C**1**. "E3.fab.local" is an address of the edge processing server **24**E**3**. "E4.mobile.local" is an address of the edge processing server **24**E**4**. "IGW1111" is an address of the terminal **15**.

(72) FIG. **8** is an explanatory diagram illustrating an example of the communication module repository **800**. The communication module repository **800** defines information on the communication module. The communication module repository **800** includes, as fields, a communication module name **801**, a description **802**, a provision form **803**, a communication form **804**, a configuration **805**, and a storage destination **806**.

(73) The communication module name **801** is a name of the communication module divided by the classification **701**. The description indicates a meaning of the communication module. The provision form **803** illustrates a form of providing the communication module. The communication form **804** indicates a form in which the communication module communicates. The configuration **805** illustrates a method in which the communication module communicates. The storage destination **806** indicates an address at which the communication module is stored.

(74) FIG. **9** is an explanatory diagram illustrating an example of the communication path requirement template table **900**. The communication path requirement template table **900** is a table that defines a template (hereinafter, the communication path requirement template) corresponding to the communication path requirement **605**. The communication path requirement template table

900 includes, as fields, a communication path requirement definition **901**, an available communication module **902**, and an available setting **903**. A combination of the communication path requirement definition **901**, the available communication module **902**, and the available setting **903** in the same row defines one communication path requirement template.
(75) The communication path requirement definition **901** defines the communication path requirement **605**. Specifically, for example, the communication path requirement definition **901** includes, as subfields, a classification **911** and a condition **912**. The classification **911** is a classification obtained by classifying communication paths by type, and includes, for example, control communication, media communication, and secure communication. The control communication is a classification **911** of the communication path for communicating the control data. The media communication is a classification **911** of the communication path for communicating the video data. The secure communication is a classification **911** of the communication paths for securely the communicating data. The condition **912** is a constraint in a case where communication classified by the classification **911** is performed.
(76) The available communication module **902** is a communication module available by the communication path requirement definition **901**. The available setting **903** is a communication setting available in the available communication module **902** by the communication path requirement definition **901**. In the available setting **903**, more detailed constraints are defined than in the condition **912**.
Exemplary Functional Configurations of Processing Server **24** and IoT Gateway **13**
(77) FIG. **10** is a block diagram illustrating a functional configuration example of the processing server **24** and the IoT gateway **13**. Each of the processing server **24** and the IoT gateway **13** includes a metric measurement unit **1001**, a module execution unit **1002**, an inter-module routing control unit **1003**, and a communication path management table **1100**.
(78) Specifically, the metric measurement unit **1001**, the module execution unit **1002**, and the inter-module routing control unit **1003** are realized, for example, by causing the processor **201** to execute a program stored in the storage device **202** illustrated in FIG. **2**. The communication path management table **1100** is stored in the storage device **202**. The metric measurement unit **1001** measures the metric information and transmits the measured metric information to the orchestration server **102**.
(79) The module execution unit **1002** executes the module arranged from the orchestration server **102**.
(80) The inter-module routing control unit **1003** executes routing control with other modules.
(81) Communication Path Management Table
(82) The communication path management table is illustrated with reference to FIGS. **11**A to **11**D.
(83) FIG. **11**A is an explanatory diagram illustrating an example of a communication path management table **1100**A retained by the edge processing server **24**E**3** in the factory LAN **32**. FIG. **11**B is an explanatory diagram illustrating an example of a communication path management table **1100**B retained by the edge processing server **24**E**4** in the local 5G network **20**. FIG. **11**C is an explanatory diagram illustrating an example of a communication path management table **1100**C retained by the IoT gateway **13**. FIG. **11**D is an explanatory diagram illustrating an example of a communication path management table **1100**D retained by the cloud processing server **24**C**1**. In a case where the communication path management tables **1100**A to **1100**D and a communication path management table **1100**E to be described later in FIG. **17**A are not distinguished, these tables are simply referred to as a communication path management table **1100**.
(84) The communication path management table **1100** includes, as fields, a management identifier **501**, a transmission source address **1101**, a destination address **1102**, and a gateway **1103**. The transmission source address **1101** is an address indicating a transmission source of a packet. The destination address **1102** is an address indicating a destination of the packet. The gateway **1103** is a gateway that is a transmission destination of the packet specified by the transmission source

address **1101** and the destination address **1102**.

(85) For example, taking the communication path management table **1100**A as an example, an entry in a first row indicates that a packet of which the transmission source is the primary learning module **405** (10.0.3.10) is sent to any destination (ANY) according to a routing table (not illustrated) of “Default”. An entry in a second row indicates that a packet of which the transmission source is the data collection module **403** (10.0.3.20) is sent to a broadband communication module (10.0.3.254) which is the gateway **1103** in a case where the camera control module **401** (10.0.5.10) is the destination and is sent to a destination (ANY) other than the camera control module **401** (10.0.5.10) according to a routing table of “Default” (not illustrated).

(86) Deployment Operation Sequence of Module

(87) FIG. **12** is a sequence diagram illustrating an example of a module deployment operation.

(88) Step S**1200**

(89) The orchestration server **102** acquires the application requirement for each application by the application requirement acquisition unit **301** and stores the application requirement in the application requirement table **600**.

(90) Step S**1201**

(91) The orchestration server **102** determines the arrangement destination of the application module **602** by the module arrangement determination unit **302** while referring to the resource management table **500** and the application requirement table **600**. Specifically, for example, the module arrangement determination unit **302** specifies the arrangement position **603** for each application module **602** while referring to the application requirement table **600**. For example, the arrangement position is “closed area” for the data collection module **403**, the video analysis module **404**, and the primary learning module **405**, the arrangement position is “field area” for the camera control module **401** and the control device control module **402**, and the arrangement position is “ANY” for the secondary learning module **406**.

(92) The module arrangement determination unit **302** specifies the position **505** corresponding to the specified arrangement position **603** while referring to the resource management table **500**. For example, in a case where the specified arrangement position **603** is “ANY”, the module arrangement determination unit **302** determines the arrangement destination of the application module **602** to be the available computer resource **504** of which the position **505** is present in either “public area” or “closed area”. In this example, the module arrangement determination unit **302** determines the arrangement destination of the secondary learning module **406** of which the arrangement position **603** is “ANY” to be the cloud processing server **24**C**1** of which the position **505** is “public area”.

(93) For example, in a case where the specified arrangement position **603** is “closed area”, the position **505** is “closed area”, and the available computer resource **504** that satisfies the communication path requirement **605** of the specified arrangement position **603** is determined to be the arrangement destination.

(94) For example, in the case of the data collection module **403** and the primary learning module **405** of which the position **505** is “closed area”, since the communication path requirement **605** is not defined, the module arrangement determination unit **302** may use any available computer resource **504** as the arrangement destination in a case where the position **505** is “closed area”. In this case, the module arrangement determination unit **302** may determine the same available computer resource **504** as the other application modules **602** to be the arrangement destination, or may determine the available computer resource **504** different from the other application modules **602** to be the arrangement destination. In this example, the module arrangement determination unit **302** determines the arrangement destination of the data collection module **403** and the primary learning module **405** to be the edge processing server **24**E**3** of which the position **505** is “closed area”.

(95) For example, in the case of the video analysis module **404** of which the position **505** is “closed

area”, the communication path requirement **605** is “control communication” and “delay <20 [ms]”. In this case, the module arrangement determination unit **302** determines the available computer resource **504** of which the position **505** is “closed area” and the delay (time) satisfies “delay <20 [ms]” of the communication path requirement **605** to be the arrangement destination. The available computer resources **504** of which the position **505** is “closed area” are the edge processing servers **24**E**2**, **24**E**3**, and **24**E**4**.

(96) Among the servers, the available computer resource **504** of which the delay (time) satisfies “delay <20 [ms]” of the communication path requirement **605** is the edge processing server **24**E**4** of the via local 5G **507**. Accordingly, the module arrangement determination unit **302** determines the arrangement destination of the video analysis module **404** of which the position **505** is “closed area” to be the edge processing server **24**E**4** of the via local 5G **507**.

(97) For example, in a case where the specified arrangement position **603** is “field area”, the module arrangement determination unit **302** determines the arrangement destination of the application modules **602** (camera control module **401** and control device control module **402**) to be the IoT gateway **13** of the terminal **15**.

(98) In the module arrangement determination unit **302**, the module **702** of the application management table **700** registers the address of the determined arrangement destination in the arrangement position address **704** of the entry which is the application module **602** of which the arrangement destination is determined.

(99) In this manner, the arrangement destination of the secondary learning module **406** is determined to be the cloud processing server **24**C**1**, the arrangement destination of the data collection module **403** and the primary learning module **405** is determined to be the edge processing server **24**E**3**, the arrangement destination of the video analysis module **404** is determined to be the edge processing server **24**E**4** of the via local 5G **507**, and the arrangement destination of the camera control module **401** and the control device control module **402** is determined to be the IoT gateway **13** of the terminal **15**.

(100) Step S**1202**

(101) The orchestration server **102** specifies the communication path requirement template corresponding to the communication path requirement **605** of the application module **602** of which the arrangement destination is determined by the module arrangement determination unit **302** from the communication path requirement template table **900** of FIG. **9** by the communication path requirement interpretation unit **303**.

(102) Specifically, for example, in the application requirement table **600** of FIG. **6**, since the communication path requirement **605** is “**-**”, the communication path requirement interpretation unit **303** does not define the communication path requirement **605** for the primary learning module **405** and the secondary learning module **406**. Accordingly, the communication path requirement interpretation unit **303** does not specify the communication path requirement template corresponding to the communication path requirement **605** for the primary learning module **405** and the secondary learning module **406**.

(103) On the other hand, in the application requirement table **600** of FIG. **6**, the communication path requirement **605** from the camera control module **401** to the data collection module **403** is defined as “media communication” and “4K video×3”. Accordingly, communication path requirement interpretation unit **303** specifies the communication path requirement templates (entry of which the classification **911** is “medial communication” and the condition **912** is “4K video×3” (N≥3)) corresponding to “media communication” and “4K video×3” from communication path requirement template table **900** of FIG. **9**.

(104) In application requirement table **600** of FIG. **6**, the communication path requirement **605** from the video analysis module **404** to the control device control module **402** is defined as “control communication” and “delay <20 [ms]”. Accordingly, the communication path requirement interpretation unit **303** specifies the communication path requirement templates (entry of which the

classification **911** is "control communication" and the condition **912** is "delay <20 [ms]" (X≥20)) corresponding to "control communication" and "delay <20 [ms]" from the communication path requirement template table **900** of FIG. **9**.
(105) In the application requirement table **600** of FIG. **6**, the communication path requirement **605** from the video analysis module **404** to the camera control module **401** is defined as "media communication" and "4K video×1". Accordingly, the communication path requirement interpretation unit **303** specifies the communication path requirement templates (entry of which the classification **911** is "media communication" and the condition **912** is "4K video×N" (N≥1)) corresponding to "media communication" and "4K video×1" from communication path requirement template table **900** of FIG. **9**.
(106) In this manner, the communication path requirement template for each communication path requirement target **604** of the application module **602** is specified.
(107) Step S**1203**
(108) The orchestration server **102** determines the available communication module **902** and the available setting **903** from the communication path requirement template specified in step S**1202** by the module arrangement determination unit **302**.
(109) Specifically, for example, since the communication path requirement **605** is not defined, the module arrangement determination unit **302** does not specify the communication path requirement template for the data collection module **403**, the primary learning module **405**, and the secondary learning module **406**. Accordingly, the module arrangement determination unit **302** does not determine the available communication module **902** and the available setting **903** for the data collection module **403**, the primary learning module **405**, and the secondary learning module **406**.
(110) In FIG. **9**, the communication path requirement template of the video analysis module **404** as the application module **602** is an entry of which the classification **911** is "control communication" and the condition **912** is "delay <20 [ms]" (X≥20). Accordingly, the module arrangement determination unit **302** determines the available communication module **902** to be the highly-reliable communication module in the communication path requirement template (entry of which the classification **911** is "control communication" and the condition **912** is "delay <20 [ms]" (X≥20)) for the video analysis module **404** as the application module **602**.
(111) The module arrangement determination unit **302** specifies the metric information **506** of the edge processing server **24**E**4** of the via local 5G **507**, which is the arrangement destination of the video analysis module **404** from the resource management table **500**. The module arrangement determination unit **302** determines the available setting **903** of the highly-reliable communication module used by the video analysis module **404** to be the specified metric information (delay: 10 [ms], jitter: 10 [ms], bandwidth 80 [Mbps], PER: 10.sup.−4 (=0.01 [%])).
(112) In this manner, the available communication module **902** in a case where the control device control module **402** and the video analysis module **404** communicate with each other is determined to be the highly-reliable communication module, and the available setting **903** is determined to be the metric information (delay: 10 [ms], jitter: 10 [ms], bandwidth 80 [Mbps], PER: 10.sup.−4 (=0.01 [%])).
(113) In FIG. **9**, the communication path requirement template of the camera control module **401** of which the communication path requirement target **604** is the data collection module **403** is an entry of which the classification **911** is "media communication" and the condition **912** is "4K video×N" (N≥3). Accordingly, the module arrangement determination unit **302** determines the available communication module **902** to be the broadband communication module in the communication path requirement template (the entry of which the classification **911** is "media communication" and the condition **912** is "4K video×N" (N≥3)) for the camera control module **401** of which the communication path requirement target **604** is the data collection module **403**.
(114) In a case where the arrangement destination of the application module **602** is the IoT device **502** (that is, the IoT gateway **13**), the module arrangement determination unit **302** specifies the

metric information **506** of the arrangement destination of the application module **602** of the communication path requirement target **604** from the resource management table **500**.
(115) In this case, the communication path requirement target **604** of the camera control module **401** is the data collection module **403**, and the available communication module **902** is determined to be the broadband communication module. Accordingly, the module arrangement determination unit **302** specifies the metric information **506** of the edge processing server **24**E**3** which is the arrangement destination of the data collection module **403** from the resource management table **500**.
(116) Specifically, for example, in a case where the edge processing server **24**E**3** communicates with the camera control module **401** of the via local 5G **507**, the module arrangement determination unit **302** specifies the metric information (delay: 30 [ms], jitter: 10 [ms], bandwidth 60 [Mbps], PER: 10.sup.−4 (=0.01 [%])) from the resource management table **500**. In a case where the edge processing server **24**E**3** communicates with the camera control module **401** of the via wireless LAN **508**, the module arrangement determination unit **302** specifies the metric information (delay: 30 [ms], jitter: 20 [ms], bandwidth 100 [Mbps], PER: 10−2 (=1 [%])) from the resource management table **500**.
(117) The module arrangement determination unit **302** determines the available setting **903** for the data collection module **403** determined to be the broadband communication module to be the specified metric information. Specifically, for example, the available setting **903** of the broadband communication module of the data collection module **403** is determined to be the metric information (delay: 30 [ms], jitter: 10 [ms], bandwidth 60 [Mbps], PER: 10.sup.−4 (=0.01 [%])) in a case where the edge processing server **24**E**3** communicates with the camera control module **401** of the via local 5G **507**, and is determined to be the metric information (delay: 30 [ms], jitter: 20 [ms], bandwidth 100 [Mbps], PER: 10.sup.−2 (=1 [%])) in a case where the edge processing server **24**E**3** communicates with the camera control module **401** of the via wireless LAN **508**.
(118) In this manner, the available communication module **902** in a case where the data collection module **403** and the camera control module **401** communicate with each other is determined to be the broadband communication module.
(119) The available setting **903** is determined to be the metric information corresponding to a communication scheme (the via local 5G **507** or the via wireless LAN **508**) between the edge processing server **24**E**3** as the arrangement destination of the data collection module **403** and the camera control module **401**.
(120) In this manner, the available communication module **902** and the available setting **903** are determined for each specified communication path requirement template.
(121) Step S**1204**
(122) The orchestration server **102** determines the arrangement destination of the available communication module **902** by the module arrangement determination unit **302** while referring to the arrangement position address **704** of the application module **602** illustrated in FIG. **7**.
(123) Specifically, for example, the module arrangement determination unit **302** determines the arrangement destination of the highly-reliable communication module which is the available communication module **902** for each of the control device control module **402** and the video analysis module **404** for the highly-reliable communication module determined as the available communication module **902** in a case where the control device control module **402** and the video analysis module **404** communicate with each other.
(124) More specifically, for example, the module arrangement determination unit **302** determines the arrangement destination of the available communication module **902** of the control device control module **402** to be "IGW1111" which is the same address as the arrangement position address **704** of the control device control module **402**, and registers the arrangement destination in the arrangement position address **704**.
(125) Similarly, the module arrangement determination unit **302** determines the arrangement

destination of the highly-reliable communication module which is the available communication module **902** of the video analysis module **404** to be “E4.mobile.local” which is the same address as the arrangement position address **704** of the video analysis module **404**, and registers the arrangement destination in the arrangement position address **704**.
(126) The module arrangement determination unit **302** determines the arrangement destination of the broadband communication module which is the available communication module **902** for each of the data collection module **403** and the camera control module **401** for the broadband communication module determined as the available communication module **902** in a case where the data collection module **403** and the camera control module **401** communicate with each other.
(127) More specifically, for example, the module arrangement determination unit **302** determines the arrangement destination of the broadband communication module which is the available communication module **902** of the data collection module **403** to be “E3.fab.local” which is the same address as the arrangement position address **704** of the data collection module **403**, and registers the arrangement destination in the arrangement position address **704**.
(128) Similarly, the module arrangement determination unit **302** determines the arrangement destination of the broadband communication module which is the available communication module **902** of the camera control module **401** to be “IGW1111” which is the same address as the arrangement position address **704** of the camera control module **401**, and registers the arrangement destination in the arrangement position address **704**.
(129) In this manner, the arrangement position address **704** of the available communication module **902** is determined.
(130) Step S**1205**
(131) The orchestration server **102** executes satisfaction confirmation of the communication path requirement **605** by the module arrangement determination unit **302**.
(132) Specifically, for example, the module arrangement determination unit **302** determines whether or not the metric information **506** determined in the available setting **903** satisfies the communication path requirement **605**, notifies a supply source of the application requirement in step S**1200** that the communication path requirement **605** is not satisfied in a case where there is the metric information **506** that does not satisfy, and deletes the entry of the application requirement registered in the application requirement table **600** and a value of the arrangement position address **704** of the application management table **700**.
(133) Step S**1206**
(134) In a case where the communication path requirement **605** is satisfied in step S**1205**, the orchestration server **102** executes the module deployment by the module deployment control unit **304**. Specifically, for example, the module deployment control unit **304** transmits the module **702** to the arrangement position address **704** according to the application management table **700** illustrated in FIG. **7**.
(135) As a result, the secondary learning module **406** is arranged in the cloud processing server **24**C**1**. The primary learning module **405** and the data collection module **403** are arranged in the edge processing server **24**E**3**. The broadband communication module which is the available communication module **902** of the data collection module **403** is also arranged in the edge processing server **24**E**3**.
(136) The video analysis module **404** is arranged in the edge processing server **24**E**4**. The highly-reliable communication module which is the available communication module **902** of the video analysis module **404** is also arranged in the edge processing server **24**E**4**.
(137) The camera control module **401** and the control device control module **402** are arranged in the IoT gateway **13**. The broadband communication module that is the available communication module **902** of the camera control module **401** is also arranged in the IoT gateway **13**. The highly-reliable communication module which is the available communication module **902** of the control device control module **402** is also placed in the IoT gateway **13**.

(138) Step S**1207**
(139) The orchestration server **102** executes a QoS setting for the factory LAN management server **35** and the mobile core apparatus **23** by the E2E network quality control unit **305**. Specifically, for example, the E2E network quality control unit **305** executes a QoS setting so as to be the provision form **803**, the communication form **804**, and the configuration **805** corresponding to the communication module while referring to the communication module repository **800**. For example, since communication between the data collection module **403** and the camera control module **401** is performed by the broadband communication module, the E2E network quality control unit **305** performs a QOS setting for each broadband communication module of the data collection module **403** and the camera control module **401** with the provision form **803** as a container, the communication form **804** as a gateway, and the configuration **805** as a point-to-point. For the content of the QoS setting, the orchestration server **102** follows the communication path Qos setting described in the available setting **903** for each communication module while referring to the communication path requirement template table **900** of FIG. **9**.
(140) Step S**1208**
(141) The orchestration server **102** executes a communication path setting for the IoT gateway **13** and the processing server **24** in which the modules are arranged by the E2E network quality control unit **305**. Specifically, for example, the E2E network quality control unit **305** sets the transmission source address **1101**, the destination address **1102**, and the gateway **1103** of the packet via the IoT gateway **13** and the processing server **24** in which the modules are arranged.
(142) For example, for the edge processing server **24**E**3**, the E2E network quality control unit **305** performs a setting such that the packet of which the transmission source address **1101** is “10.0.3.10” which is the module address **703** of the primary learning module **405** is sent according to a routing table (not illustrated) of “Default” to any destination address **1102** (ANY).
(143) The E2E network quality control unit **305** performs a setting such that the packet of which the transmission source address **1101** is “10.0.3.20” which is the module address **703** of the data collection module **403** is sent to the broadband communication module (10.0.3.254) which is the gateway **1103** in a case where the module address **703** “10.0.5.10” of the camera control module **401** is the destination address **1102** and is sent according to a routing table (not illustrated) of the gateway **1103** of “Default” for the destination address **1102** (ANY) other than the camera control module **401** (10.0.5.10).
(144) In this manner, a logical system by the deployment of the module is constructed.
(145) Step S**1209**A
(146) The IoT gateway **13** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the IoT gateway **13** starts executing the camera control module **401**, the broadband communication module thereof, the control device control module **402**, and the highly-reliable communication module thereof.
(147) Step S**1209**B
(148) The edge processing server **24**E**4** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the edge processing server **24**E**4** starts executing the video analysis module **404** and the highly-reliable communication module.
(149) Step S**1209**C
(150) The edge processing server **24**E**3** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the edge processing server **24**E**3** starts executing the data collection module **403**, the highly-reliable communication module thereof, and the primary learning module **405**.
(151) Step S**1209**D
(152) The cloud processing server **24**C**1** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the cloud processing server **24**C**1** starts executing the secondary learning module **406**.

(153) As a result, an operation of a logical system constructed by the deployment of the module is started.
(154) Logical System after Deployment
(155) FIG. **13** is a block diagram illustrating a configuration example of a logical system after the deployment of the module. FIG. **13** illustrates a logical system constructed according to the sequence of FIG. **12**. In FIG. **13**, a bidirectional arrow between the modules indicates that both the modules can communicate with each other.
(156) Communication between the data collection module **403** and the camera control module **401** is realized by broadband communication **1301**T between a broadband communication module **1301**B connected to the data collection module **403** and a broadband communication module **1301**A connected to the camera control module **401**. Communication between the video analysis module **404** and the control device control module **402** is realized by highly-reliable communication **1302**T between a highly-reliable communication module **1302**B connected to the video analysis module **404** and a highly-reliable communication module **1302**A connected to the control device control module **402**.
Display Screen Example of Orchestration Server **102**
(157) FIG. **14** is an explanatory diagram illustrating a display screen example of the orchestration server **102**. A display screen **1400** of FIG. **14** displays, as a deployment processing result, the storage contents of the application management table **700** illustrated in FIG. **7**.
(158) Undeployment Operation Sequence of Application
(159) FIG. **15** is a sequence diagram illustrating an example of an undeployment operation of the module. The undeployment operation of the module of FIG. **15** indicates an operation after the deployment operation of the module of FIG. **12**.
(160) Step S**1500**
(161) The orchestration server **102** acquires an application deletion request by the application requirement acquisition unit **301**.
(162) Step S**1501**
(163) The orchestration server **102** transmits a deletion instruction of the communication path setting to the IoT gateway **13** and the processing server **24** for which the communication path setting is performed (step S**1208**) by the E2E network quality control unit **305**.
(164) Step S**1502**
(165) The orchestration server **102** transmits a deletion instruction of the QOS setting to the factory LAN management server **35** and the mobile core apparatus **23** that have performed the QOS setting (step S**1207**) by the E2E network quality control unit **305**. The factory LAN management server **35** deletes the QOS setting, and thus, communication between the edge processing server **24**E**3** and the IoT gateway **13** via the factory LAN **32** is interrupted. The mobile core apparatus **23** deletes the QOS setting, and thus, communication between the edge processing server **24**E**4** and the IoT gateway **13** via the local 5G network **20** is interrupted.
(166) Step S**1503**
(167) The orchestration server **102** transmits a deletion instruction of the module to the IoT gateway **13** and the processing server **24** in which the modules are arranged by the E2E network quality control unit **305**.
(168) Step S**1504**A
(169) Upon receiving the deletion instruction in step S**1501**, the IoT gateway **13** deletes the entry of the management identifier **501** “IGW1111” serving as a deletion instruction target in the communication path management table **1100**C. Upon receiving the deletion instruction in step S**1503**, the IoT gateway **13** deletes the arranged camera control module **401**, the broadband communication module **1301**A thereof, the control device control module **402**, and the highly-reliable communication module **1302**A thereof.
(170) Step S**1504**B

(171) Upon receiving the deletion instruction in step S**1501**, the edge processing server **24**E**4** deletes the entry of the management identifier **501** "IGW1111" serving as a deletion instruction target of the communication path management table **1100**B. Upon receiving the deletion instruction in step S**1503**, the edge processing server **24**E**4** deletes the arranged video analysis module **404** and the highly-reliable communication module **1302**B.
(172) Step S**1504**C
(173) Upon receiving the deletion instruction in step S**1501**, the edge processing server **24**E**3** deletes the entry of the management identifier **501** "IGW1111" serving as a deletion instruction target of the communication path management table **1100**A. Upon receiving the deletion instruction in step S**1503**, the edge processing server **24**E**3** deletes the arranged data collection module **403**, the highly-reliable communication module thereof, and the primary learning module **405**.
(174) Step S**1504**D
(175) Upon receiving the deletion instruction in step S**1501**, the cloud processing server **24**C**1** deletes the entry of the management identifier "IGW1111" serving as a deletion instruction target of the communication path management table **1100**D. Upon receiving the deletion instruction in step S**1503**, the cloud processing server **24**C**1** deletes the arranged secondary learning module **406**.
(176) In this manner, the module is undeployed from the processing server **24**.
(177) Re-Deployment Operation Sequence of Module
(178) FIG. **16** is a sequence diagram illustrating an example of a re-deployment operation of the module. The re-deployment is executed, for example, in a case where there is a change in the metric information **506**. In FIG. **16**, an example in which the video analysis module **404** and the highly-reliable communication module **1302**B thereof are re-arranged in the edge processing server **24**E**2** in a case where the edge processing server **24**E**4** no longer satisfies the communication path requirement **605** of the video analysis module **404** which is the application module **602** arranged in the edge processing server **24**E**4** will be described.
(179) Step S**1600**
(180) The orchestration server **102** detects the application module **602** that does not satisfy the communication path requirement **605** due to the change in the metric information **506** by the metric information collection unit **300**. Specifically, for example, the metric information collection unit **300** repeatedly collects the metric information **506** for each available computer resource **504**, and detects whether the communication path requirement **605** of the application module **602** arranged in the available computer resource **504** is no longer satisfied.
(181) In this example, the metric information collection unit **300** detects that the edge processing server **24**E**3** no longer satisfies the communication path requirement **605** (4K video×3) of the data collection module **403** arranged in the edge processing server **24**E**3**.
(182) Step S**1601**
(183) In a case where the orchestration server **102** detects the application module **602** that does not satisfy the communication path requirement **605** in step S**1600**, the orchestration server **102** determines a re-arrangement destination of the application module **602** by the module arrangement determination unit **302** while referring to the resource management table **500** and the application requirement table **600** as in step S**1201**. In this example, a re-arrangement destination of the data collection module **403** is determined to be the edge processing server **24**E**2**.
(184) Steps S**1602** to S**1605**
(185) Steps S**1602** to S**1605** are the same processing as steps S**1202** to S**1205**. In this example, a re-arrangement destination of the broadband communication module **1301**B of the data collection module **403** is determined to be the edge processing server **24**E**2** in steps S**1602** to S**1605**. Since the primary learning module **405** communicates with the data collection module **403** in the edge processing server **24**E**3**, the orchestration server **102** may be re-arranged in the edge processing server **24**E**2** together with the data collection module **403** and the broadband communication module **1301**B.

(186) Step S**1606**
(187) As in step S**1501**, the orchestration server **102** transmits a deletion instruction of the communication path setting to the edge processing server **24**E**3** which is an original arrangement destination of the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405**.
(188) Step S**1607**
(189) As in step S**1503**, the orchestration server **102** transmits a deletion instruction of the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405** to the edge processing server **24**E**3** which is an original arrangement destination of the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405**.
(190) Step S**1608**
(191) Upon receiving the deletion instruction in step S**1606**, the edge processing server **24**E**3** deletes the entry of the management identifier **501** "IGW1111" serving as a deletion instruction target of the communication path management table **1100**A. Upon receiving the deletion instruction in step S**1607**, the edge processing server **24**E**3** deletes the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405**.
(192) Step S**1607**
(193) The orchestration server **102** executes module deployment for the IoT gateway **13**, the edge processing server **24**E**2**, the edge processing server **24**E**4**, and the cloud processing server **24**C**1** according to the processing results of steps S**1601** to S**1605** by the module deployment control unit **304**.
(194) In this manner, a logical system by the re-deployment of the module is constructed.
(195) Step S**1610**A
(196) The IoT gateway **13** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the IoT gateway **13** starts executing the camera control module **401**, the broadband communication module **1301**A, the control device control module **402**, and the highly-reliable communication module **1302**A.
(197) Step S**1610**B
(198) The edge processing server **24**E**4** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the edge processing server **24**E**2** starts executing the video analysis module **404** and the highly-reliable communication module **1302**B.
(199) Step S**1610**D
(200) The cloud processing server **24**C**1** starts executing the module arranged from the orchestration server **102**. Specifically, for example, the cloud processing server **24**C**1** starts executing the secondary learning module **406**.
(201) Step S**1610**E
(202) The edge processing server **24**E**2** starts executing the module arranged from the orchestration server **102** in step S**1609**. Specifically, for example, the edge processing server **24**E**3** starts executing the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405**.
(203) As a result, an operation of a logical system constructed by the re-deployment of the module is started.
(204) Communication Path Management Table **1100** after Re-Deployment
(205) The communication path management table **1100** after re-deployment will be described with reference to FIGS. **17**A and **17**B.
(206) FIG. **17**A is an explanatory diagram illustrating an example of a communication path management table **1100**E after re-deployment retained by the edge processing server **24**E**2** in the public 5G network **40**. FIG. **17**B is an explanatory diagram illustrating an example of the communication path management table **1100**C after re-deployment retained by the IoT gateway **13**.

(207) As illustrated in FIG. **17**A, after the re-deployment, the communication path management table **1100**E of the edge processing server **24**E**2** is set instead of the communication path management table **1100**A of the edge processing server **24**E**3**.
(208) In the communication path management table **1100**E of the edge processing server **24**E**2** illustrated in FIG. **17**A, an entry in a first row indicates that a packet of which the transmission source is the primary learning module **405** (10.0.2.10) is sent to the gateway **1103** of “Default” for any destination (ANY). An entry in a second line indicates that a packet of which the transmission source is the data collection module **403** (10.0.3.20) is sent to the broadband communication module (10.0.3.254) which is the gateway **1103** in a case where the camera control module **401** (10.0.5.10) is the destination and is sent to the gateway **1103** of “Default” for a destination (ANY) other than the camera control module **401** (10.0.5.10).
(209) As illustrated in FIG. **17**B, in the communication path management table **1100**C of the IoT gateway **13**, the address of the primary learning module **405** in a shaded portion of the transmission destination address **1103**C is changed from “10.0.3.20” to “10.0.2.20” as compared with FIG. **11**.
(210) The communication path management table **1100**B retained by the edge processing server **24**E**4** and the communication path management table **1100**D retained by the cloud processing server **24**C**1** are not changed from the tables in FIGS. **11**B and **11**D even after re-deployment.
(211) Logical System after Re-Deployment
(212) FIG. **18** is a block diagram illustrating a configuration example of a logical system after the re-deployment of the module. The difference from FIG. **13** is that the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405** are re-arranged from the edge processing server **24**E**3** of the factory LAN **32** to the edge processing server **24**E**2** of the public 5G network **40**.
Display Screen Example of Orchestration Server **102** after Re-Deployment
(213) FIG. **19** is an explanatory diagram illustrating a display screen example of the orchestration server **102** after re-deployment. As compared with FIG. **14**, the module address **703** and the arrangement position address **704** of the broadband communication module **1301**B and the primary learning module **405** are rewritten to the address of the edge processing server **24**E**2** as the re-deployment destination. The module address **703** and the arrangement position address **704** of the application management table **700** are updated as illustrated in FIG. **19**.
(214) In the re-deployment described above, although it has been described that the data collection module **403**, the broadband communication module **1301**B, and the primary learning module **405** are re-arranged from the edge processing server **24**E**3** of the factory LAN **32** to the edge processing server **24**E**2** of the public 5G network **40**, the broadband communication module **1301**A, the camera control module **401**, the control device control module **402**, and the highly-reliable communication module **1302**A may be re-arranged in another IoT gateway **13**.
(215) As described above, according to the present embodiment, for example, it is possible to realize highly-reliable connectivity (communication path) such that a site network in a site such as manufacturing or distribution satisfies the application requirement requested by the application.
(216) The present invention is not limited to the aforementioned embodiment, and includes various modification examples and equivalent configurations within the gist of the appended claims. For example, the aforementioned embodiment is described in detail in order to facilitate easy understanding of the present invention, and the present invention is not limited to necessarily including all the described components. A part of the configuration of one embodiment may be replaced with the configuration of another embodiment. The configuration of another embodiment may be added to the configuration of one embodiment. Another configuration may be added, removed, and substituted to, from, and into some of the configurations of the aforementioned embodiment.
(217) A part or all of the aforementioned configurations, functions, processing units, and processing means may be realized by hardware by being designed with, for example, an integrated

circuit. Alternatively, the processor interprets and executes a program for realizing the functions, and thus, a part or all of the aforementioned configurations, functions, processing units, and processing means may be realized by software.
(218) Information of programs, tables, and files for realizing the functions can be stored in a storage device such as a memory, a hard disk, or a solid state drive (SSD), or a recording medium such as an integrated circuit (IC) card, an SD card, or a digital versatile disc (DVD).
(219) Control lines and information lines illustrate lines which are considered to be necessary for the description, and not all the control lines and information lines necessary in the implementation are necessarily illustrated. Almost all the configurations may be considered to be actually connected to each other.

Claims

1. A management apparatus that is able to communicate with a computer group, the apparatus comprising: a processor that executes a program; and a storage device that stores the program, wherein the storage device stores an application module group, a communication module group, a communication path requirement that is a condition required for a communication path between an application module in the application module group and another application module as a communication partner of the application module, a communication path requirement target that designates the other application module, and a communication path requirement template group that includes a communication path requirement definition that defines a type and a condition of communication in the communication path requirement, an available communication module that defines a communication module available to the communication path requirement definition, and an available setting that defines a setting of communication available to the communication path requirement definition, and the processor executes specification processing of specifying a first communication path requirement template corresponding to a first communication path requirement of a first application module from the communication path requirement template group, and specifying a second communication path requirement template corresponding to a second communication path requirement of a second application module designated as a communication partner of the first application module by the communication path requirement target from the communication path requirement template group, communication module determination processing of determining the available communication module included in the first communication path requirement template to be a first communication module available by the first application module, and determining the available communication module included in the second communication path requirement template to be a second communication module available by the second application module, communication module arrangement destination determination processing of determining an arrangement destination of the first communication module to be a first arrangement destination of the first application module in the computer group, and determining an arrangement destination of the second communication module to be a second arrangement destination of the second application module in the computer group, arrangement processing of arranging the first application module and the first communication module in the first arrangement destination and arranging the second application module and the second communication module in the second arrangement destination, and communication path setting processing of setting a communication path that connects the first application module and the second application module to be able to communicate via the first communication module and the second communication module.
2. The management apparatus according to claim 1, wherein the storage device stores area information on a position of each of computers of the computer group, and an arrangeable area of each application module of the application module group, and the processor executes application module arrangement destination determination processing of determining the first arrangement

destination from the computer group, and determining the second arrangement destination from the computer group based on the area information and the arrangeable area.

3. The management apparatus according to claim 1, wherein the storage device stores a communication module repository that defines information on network quality corresponding to a type of the communication module, and the processor executes quality setting processing of setting network quality corresponding to a type of each communication module of the first communication module and the second communication module while referring to the communication module repository.

4. The management apparatus according to claim 1, wherein the processor executes deletion processing of transmitting a deletion instruction of the communication path to the first arrangement destination and the second arrangement destination, transmitting a stop or deletion instruction of the first application module and the first communication module to the first arrangement destination, and transmitting a stop or deletion instruction of the second application module and the second communication module to the second arrangement destination.

5. The management apparatus according to claim 2, wherein the storage device stores metric information generated by communication between the computers, and the processor executes first detection processing of detecting that the first application module does not satisfy the first communication path requirement based on the metric information, application module arrangement destination re-determination processing of determining a third arrangement destination different from the first arrangement destination from the computer group based on the area information and the arrangeable area for the first application module that does not satisfy the first communication path requirement in the first detection processing, communication module arrangement destination re-determination processing of determining the arrangement destination of the first communication module to be the third arrangement destination, re-arrangement processing of arranging the first application module and the first communication module in the third arrangement destination, and communication path re-setting processing of setting a communication path that connects the first application module arranged in the third arrangement destination and the second application module to be able to communicate via the first communication module arranged in the third arrangement destination and the second communication module.

6. The management apparatus according to claim 2, wherein the storage device stores metric information generated by communication between the computers, and the processor executes second detection processing of detecting that the second application module does not satisfy the second communication path requirement based on the metric information, application module arrangement destination re-determination processing of determining a fourth arrangement destination different from the second arrangement destination from the computer group based on the area information and the arrangeable area for the second application module that does not satisfy the second communication path requirement in the second detection processing, communication module arrangement destination re-determination processing of determining the arrangement destination of the second communication module to be the fourth arrangement destination, re-arrangement processing of arranging the second application module and the second communication module in the fourth arrangement destination, and communication path re-setting processing of setting a communication path that connects the first application module and the second application module arranged in the fourth arrangement destination to be able to communicate via the first communication module and the second communication module re-arranged in the fourth arrangement destination.

7. The management apparatus according to claim 5, wherein the processor executes second detection processing of detecting that the second application module does not satisfy the second communication path requirement based on the metric information, in the application module arrangement destination re-determination processing, the processor determines a fourth arrangement destination different from the second arrangement destination from the computer

group based on the area information and the arrangeable area for the second application module that does not satisfy the second communication path requirement in the second detection processing, in the communication module arrangement destination re-determination processing, the processor determines the arrangement destination of the second communication module to the fourth arrangement destination, in the re-arrangement processing, the processor arranges the second application module and the second communication module in the fourth arrangement destination, and in the communication path re-setting processing, the processor sets a communication path that connects the first application module re-arranged in the third arrangement destination and the second application module re-arranged in the fourth arrangement destination to be able to communicate via the first communication module re-arranged in the third arrangement destination and the second communication module re-arranged in the fourth arrangement destination.

8. A management system that includes a computer group and a management apparatus that is able to communicate with the computer group, wherein the management apparatus stores an application module group, a communication module group, a communication path requirement that is a condition required for a communication path between an application module in the application module group and another application module as a communication partner of the application module, a communication path requirement target that designates the other application module, and a communication path requirement template group that includes a communication path requirement definition that defines a type and a condition of communication in the communication path requirement, an available communication module that defines a communication module available to the communication path requirement definition, and an available setting that defines a setting of communication available to the communication path requirement definition, and the management apparatus executes specification processing of specifying a first communication path requirement template corresponding to a first communication path requirement of a first application module from the communication path requirement template group, and specifying a second communication path requirement template corresponding to a second communication path requirement of a second application module designated as a communication partner of the first application module by the communication path requirement target from the communication path requirement template group, communication module determination processing of determining the available communication module included in the first communication path requirement template to be a first communication module available by the first application module, and determining the available communication module included in the second communication path requirement template to be a second communication module available by the second application module, communication module arrangement destination determination processing of determining an arrangement destination of the first communication module to be a first arrangement destination of the first application module in the computer group, and determining an arrangement destination of the second communication module to be a second arrangement destination of the second application module in the computer group, arrangement processing of arranging the first application module and the first communication module in the first arrangement destination and arranging the second application module and the second communication module in the second arrangement destination, and communication path setting processing of setting a communication path that connects the first application module and the second application module to be able to communicate via the first communication module and the second communication module.

9. A management method by a management apparatus that is able to communicate with a computer group, wherein the management apparatus includes a processor that executes a program, and a storage device that stores the program, the storage device stores an application module group, a communication module group, a communication path requirement that is a condition required for a communication path between an application module in the application module group and another application module as a communication partner of the application module, a communication path requirement target that designates the other application module, and a communication path

requirement template group that includes a communication path requirement definition that defines a type and a condition of communication in the communication path requirement, an available communication module that defines a communication module available to the communication path requirement definition, and an available setting that defines a setting of communication available to the communication path requirement definition, and the processor executes specification processing of specifying a first communication path requirement template corresponding to a first communication path requirement of a first application module from the communication path requirement template group, and specifying a second communication path requirement template corresponding to a second communication path requirement of a second application module designated as a communication partner of the first application module by the communication path requirement target from the communication path requirement template group, communication module determination processing of determining the available communication module included in the first communication path requirement template to be a first communication module available by the first application module, and determining the available communication module included in the second communication path requirement template to be a second communication module available by the second application module, communication module arrangement destination determination processing of determining an arrangement destination of the first communication module to be a first arrangement destination of the first application module in the computer group, and determining an arrangement destination of the second communication module to be a second arrangement destination of the second application module in the computer group, arrangement processing of arranging the first application module and the first communication module in the first arrangement destination and arranging the second application module and the second communication module in the second arrangement destination, and communication path setting processing of setting a communication path that connects the first application module and the second application module to be able to communicate via the first communication module and the second communication module.